

Supplementary material for “A local glyph process suffices: an information-theoretic mechanism comparison for the Voynich manuscript”

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Prepared 23 September 2026. All tables are generated from the analysis logs by the same script that builds the main text.

S1. Parsing of the IVTFF transliteration

Page headers of the form `<f1r> <! $Q=A $P=A $F=a $B=1 $I=T $L=A $H=1 ...>` supply the metadata. Locus lines `<f1r.1,@P0>` text carry a locus type whose first letter, after stripping the position markers `@ + * = ~ ! / &`, gives the kind: P (paragraph text), L (label), C (circular), R (radial). Inline markup: `[a:o]` alternatives are resolved to the first reading; `@nnn`; extended glyph codes and `?` mark a token as uncertain and such tokens are removed; `<! ... >` comments and `{...}` annotations are removed; `<->` and `<~>` drawing intrusions are treated as spaces; `<%>` and `<$>` paragraph markers are removed; `!` fillers are deleted; `.` is a space and `,` (uncertain space) is either joined (baseline) or treated as a space. The EVA alphabet is used as is; ligature and capitalized variants are kept as distinct glyphs. Resulting corpus: 4,130 paragraph-text lines on 207 pages, 32,428 tokens after removing uncertain tokens; 1,114 label tokens.

S2. Additional tables

Table S1. Share of adjacent inner tokens that share the given feature, observed / within-line permutation null = ratio.

Corpus	n pairs	first glyph	first two	last glyph	last two	last three	identical
Voynich	19,895	0.193 / 0.197 = 0.98	0.120 / 0.124 = 0.97	0.311 / 0.314 = 0.99	0.168 / 0.163 = 1.03	0.093 / 0.091 = 1.02	0.011 / 0.011 = 0.98
Currier A	5,661	0.210 / 0.212 = 0.99	0.125 / 0.125 = 1.00	0.253 / 0.251 = 1.01	0.132 / 0.130 = 1.02	0.062 / 0.061 = 1.01	0.012 / 0.012 = 0.94
Currier B	13,980	0.186 / 0.191 = 0.98	0.118 / 0.124 = 0.95	0.335 / 0.339 = 0.99	0.183 / 0.178 = 1.03	0.106 / 0.103 = 1.03	0.010 / 0.011 = 0.94
Latin	22,441	0.055 / 0.063 = 0.87	0.010 / 0.016 = 0.62	0.150 / 0.135 = 1.10	0.044 / 0.041 = 1.07	0.008 / 0.012 = 0.68	0.000 / 0.002 = 0.00
Italian	22,441	0.046 / 0.061 = 0.76	0.006 / 0.014 = 0.45	0.185 / 0.159 = 1.16	0.013 / 0.017 = 0.76	0.001 / 0.006 = 0.20	0.000 / 0.005 = 0.09
German	19,859	0.051 / 0.065 = 0.79	0.011 / 0.018 = 0.60	0.120 / 0.124 = 0.97	0.040 / 0.046 = 0.86	0.008 / 0.016 = 0.48	0.002 / 0.008 = 0.24
Hebrew	11,373	0.105 / 0.129 = 0.81	0.025 / 0.035 = 0.73	0.114 / 0.119 = 0.95	0.022 / 0.029 = 0.74	0.007 / 0.014 = 0.48	0.001 / 0.006 = 0.24

Table S2. Single-glyph edits between adjacent inner tokens at edit distance 1 (739 pairs), forward versus inverse counts; $z = (n_{ab} - n_{ba}) / \sqrt{(n_{ab} + n_{ba})}$. Share of pairs in edits whose inverse is less than half as frequent: 0.18.

edit a → b	count a → b	count b → a	z
ins t @mid	14	6	1.8
sub r>l @end	36	23	1.7
del d @mid	30	20	1.4
ins i @mid	16	10	1.2
del e @mid	50	40	1.1

sub c>s @start	17	13	0.7
del o @start	18	14	0.7
sub t>k @mid	27	24	0.4
ins q @start	37	35	0.2

Table S3. Similarity of a token to candidate copy sources (Timm and Schinner 2020) and to controls.

Relation	pairs	mean edit distance	P(identical)	P(ED ≤ 1)	P(ED ≤ 2)
prev word	30,756	4.44	0.009	0.046	0.126
word above (i-1..i + 1, min)	31,008	3.42	0.026	0.108	0.280
word above (same index)	29,512	4.50	0.010	0.044	0.125
random same page	34,859	4.61	0.012	0.038	0.107
random same lang	34,859	4.75	0.005	0.025	0.085
2 lines above (same index)	27,584	4.54	0.010	0.043	0.119
min over 3 random same page	33,042	3.34	0.037	0.109	0.280

Table S4. h_2/h_1 after a given number of byte-pair merges (inventory size in parentheses).

merges	VMS	de	it	la	he
0	0.54 (27)	0.77 (31)	0.79 (35)	0.82 (24)	0.85 (28)
5	0.58 (32)	0.78 (36)	0.79 (40)	0.82 (29)	0.84 (33)
10	0.63 (37)	0.77 (41)	0.78 (45)	0.80 (34)	0.83 (38)
15	0.66 (42)	0.76 (46)	0.78 (49)	0.79 (39)	0.82 (43)
20	0.68 (47)	0.76 (51)	0.76 (54)	0.78 (44)	0.82 (48)
25	0.69 (52)	0.75 (56)	0.76 (59)	0.77 (49)	0.81 (53)
30	0.70 (56)	0.75 (61)	0.76 (64)	0.76 (54)	0.80 (58)
40	0.72 (66)	0.74 (71)	0.75 (74)	0.75 (64)	0.79 (68)
50	0.73 (76)	0.74 (81)	0.75 (84)	0.75 (74)	0.79 (78)
60	0.73 (86)	0.73 (91)	0.75 (94)	0.74 (84)	0.78 (88)
80	0.74 (105)	0.74 (110)	0.74 (114)	0.72 (104)	0.77 (108)

Table S5. Decomposition of the adjacent-token excess MI (global marginal-preserving null) by edit distance between the two tokens.

Corpus	total excess	ED = 0	ED = 1	ED = 2	ED ≥ 3
VMS raw	0.203	0.027	0.105	0.146	-0.075
VMS light	0.309	0.041	0.079	0.135	0.055
VMS heavy	0.292	0.052	0.062	0.114	0.064
VMS-A light	0.150	0.040	0.080	0.150	-0.120
VMS-B light	0.299	0.029	0.060	0.115	0.095
Latin	0.262	-0.003	-0.000	0.022	0.243
Latin stem4	0.347	-0.003	-0.001	0.022	0.329
Italian	0.406	0.002	0.014	0.126	0.265
German	0.210	0.009	-0.002	0.021	0.182

Table S6. Page specificity of tokens by position in the line (equal n per position).

Position in line	Voynich I(token; page)	Voynich types/n	Latin I(token; page)	Italian I(token; page)
first	0.098	0.446	0.011	0.014
second	0.134	0.332	0.006	0.006
middle	0.113	0.347	0.014	0.017
penult	0.106	0.361	0.018	0.003
last	0.055	0.437	0.017	0.001

Table S7. Excess MI between tokens of consecutive lines of the same page, by position class.

Pair of consecutive lines	Voynich	Latin	Italian
mid_k -> mid_k + 1	0.013	-0.002	-0.006
last_k -> first_k + 1	0.003	0.065	0.178
mid_k -> first_k + 1	0.009	-0.007	0.006
last_k -> mid_k + 1	0.007	-0.002	0.002
first_k -> first_k + 1	0.018	0.003	0.004
last_k -> last_k + 1	0.022	-0.007	-0.002

Table S8. Page-drift automata: order o , mixing weight λ of page-level counts against section-level counts.

Automaton	edge MI	token MI	line-break MI	space gain	hapax	Zipf	attested real	attested synth.	h(2)	h(4)	residual	I(token; page)
hier o3 lam0.5	0.153	0.040	-0.003	73.7%	0.70	-0.78	0.80	0.75	1.92	1.75	+0.011	0.187
hier o4 lam0.3	0.160	0.066	0.002	73.5%	0.64	-0.74	0.81	0.75	1.93	1.70	-0.036	0.157
hier o4 lam0.5	0.158	0.076	0.004	73.8%	0.63	-0.74	0.81	0.75	1.92	1.70	-0.052	0.215
hier o4 lam0.7	0.164	0.065	-0.002	74.3%	0.64	-0.74	0.81	0.75	1.92	1.69	-0.084	0.279
hier o5 lam0.3	0.162	0.104	-0.001	73.9%	0.55	-0.73	0.83	0.74	1.92	1.70	-0.110	0.163

Table S9. Naibbe cipher variants: one card per letter or per token; with and without the 3% random space removal; independent sentences per line or running text; Latin or Italian plaintext.

Variant	edge MI	token MI	line-break MI	space gain	hapax	Zipf	attested real	attested synth.	h(2)	h(4)	residual	I(token; page)
naibbe indep, card/letter	0.003	0.055	0.002	77.4%	0.49	-1.00	0.85	0.72	1.90	1.70	-0.065	0.007
naibbe indep, card/token	0.003	0.120	0.004	77.5%	0.44	-0.75	0.95	0.53	1.88	1.60	-0.198	-0.001
naibbe indep, no merge	0.002	0.069	0.009	80.8%	0.41	-1.00	0.88	0.73	1.87	1.68	-0.074	-0.010
naibbe running	0.002	0.078	0.023	77.0%	0.49	-1.01	0.84	0.72	1.91	1.71	-0.050	-0.007
naibbe running Italian	0.002	0.066	0.029	76.8%	0.47	-1.01	0.86	0.69	1.92	1.73	-0.065	0.001

Table S10. Herbal-A: Mantel test for each single image feature.

Image feature	Mantel r (tokens)	p
green	-0.102	0.98
blue	-0.049	0.75
red	+ 0.000	0.49
green_y	-0.085	0.94
green_x	-0.006	0.54
red_y	+ 0.035	0.22
green_hue	-0.020	0.64
blue_frac	-0.059	0.80
red_frac	-0.014	0.57
colour_total	-0.120	0.99

Table S11. Additional battery columns for all mechanisms (d2_mi: tokens at distance 2, global null; break_edge / adj_edge: glyph-level (last, first) MI across the line break and for adjacent pairs at the same n; h3_resid: h(3) minus twin).

Mechanism	tokens	types	wlen	d2_mi	adj_mi_same_n	break_edge	adj_edge_same_n	h3	lzma	h3_resid
VMS	32399	7832	5.336	0.066	0.051	0.012	0.144	1.821	2.399	-0.002
markov3 + drift	35102	8397	5.276	0.026	0.030	-0.001	0.173	1.822	2.436	0.012
markov4 + drift	33366	7307	5.272	0.028	0.025	-0.002	0.152	1.820	2.377	0.009
latin	32399	7924	6.063	0.033	0.079	0.027	0.043	2.101	2.592	0.016
latin rule-sp	32399	13490	6.189	0.028	0.071	0.144	0.136	2.242	2.652	0.019
latin indep lines	31123	5669	5.850	0.532	0.204	0.012	0.037	2.057	1.992	0.008
rugg grille	32399	10637	5.346	0.597	0.157	0.001	0.001	2.343	2.252	0.031
timm-schinner v2	32399	15569	6.363	0.106	0.042	0.005	0.031	2.372	1.880	0.022
verbose + indep	44081	4573	4.438	0.728	0.411	0.006	0.422	1.786	1.530	-0.007
autokey 1g/letter	28789	18405	5.755	0.190	0.067	0.003	0.655	2.593	3.490	0.131
autokey 2g/letter	28162	14096	5.968	0.205	0.123	0.001	0.625	1.809	2.673	0.043
naibbe indep, card/letter	31562	5978	5.409	0.008	0.014	0.003	0.005	1.806	2.413	
naibbe indep, card/token	31558	2435	5.385	0.018	0.041	-0.002	0.006	1.746	2.232	
naibbe running	31552	6054	5.381	-0.003	0.021	0.004	0.001	1.817	2.437	
naibbe indep, no merge	32394	5357	5.271	0.006	0.028	0.005	0.008	1.781	2.381	
naibbe running Italian	31552	5894	5.387	0.001	0.023	0.006	-0.000	1.830	2.442	
naibbe stateful w = 0.9 c = 0.15	31526	5815	5.521	0.009	0.080	0.005	0.071	1.746	2.270	
naibbe stateful w = 1.0 c = 0.05	31584	5458	5.451	0.035	0.189	-0.001	0.253	1.686	2.142	

naibbe stateful w = 0.6 c = 0.15	31536	6113	5.476	0.010	0.033	0.001	0.034	1.795	2.366
hier o4 lam0.3	32937	7321	5.268	0.017	0.033	0.004	0.160	1.829	2.383
hier o4 lam0.5	33140	7236	5.258	0.027	0.036	0.004	0.172	1.828	2.370
hier o4 lam0.7	33154	7263	5.282	0.031	0.036	0.003	0.160	1.820	2.330
hier o3 lam0.5	34752	8143	5.269	0.028	0.030	0.006	0.162	1.823	2.420
hier o5 lam0.3	31717	6438	5.260	0.029	0.047	0.005	0.184	1.825	2.321

S3. Generator parameters

Glyph automata. Counts of order- k contexts over the glyph stream of each line with k start symbols and an end symbol; sampling per line with the target glyph length of the corresponding real line, the end symbol suppressed until 60% of the target length and sampling stopped at 140%. Section drift: one model per (Currier language, illustration section) group with at least 30 lines, otherwise the global model. Page drift: the next-symbol distribution is λ times the page's own normalized counts plus $(1 - \lambda)$ times the section's, for contexts seen on the page. Twins for the residual: order 3, section drift, fitted to the text under test.

Table and grille. Slot decomposition of each attested token by the regular expression prefix = $q?(o|y|d|s|ch|sh)?(k|t|p|f|ck|ct|cp|cf|ch|sh|ckh|cth|cph|cfh|l|r|d|s)?$, midfix = $e\{0,3\}(o|a|y)?$, suffix = $(i\{0,3\}(n|r|l|m)|d|l|r|s|y|dy|ly|ry|dl|ol|al|or|ar|am|an)?y?$, with unmatched tokens split into thirds. Per section group a table of 40 columns and $\max(2, \text{round}(0.9 \times \text{tokens}/40))$ rows, each cell an independent draw of prefix, midfix and suffix from the empirical frequencies. Each page starts at a random cell; a line is the run of cells to the right (wrapping); each line advances one row.

Copy and modify. Seeds: the 300 most frequent real tokens. Source choice: line above, same column ± 1 (0.6); previous token in the line (0.2); random token of the page (0.15); seed (rest). Modification with probability 0.85; a second modification with probability 0.25. Modifications: substitution within the similarity pairs (ch,sh), (k,t), (t,p), (k,f), (e,ee), (ee,eee), (i,ii), (ii,iii), (a,o), (y,dy), (r,l), (n,m), (ol,or), (ar,al), (ain,aiin), (ey,edy) with probability 0.5; otherwise addition or removal of a prefix from $\{q,o,d,s,y,l,ch,sh\}$ (0.25) or a suffix from $\{y,dy,n,in,iin,r,l,s,m\}$ (0.25), the choice between addition and removal depending on the current length so that tokens stay around five to six glyphs.

Verbose cipher with fixed homophone choice. Each of 23 Latin letters is assigned one to three strings from a pool of 35 Voynich-like units; the homophone is selected by the code of the previous glyph modulo the number of homophones. Spaces are inserted after y, n, r, l, m, s once the current token has three glyphs, and forced at nine. One independent Latin sentence (split at . ; : ? !) per line, truncated to the required number of tokens.

Ciphertext-feedback substitution. $c_t = T[c_{t-1}][p_t]$ with 27 rows, each a random injective map from 23 letters to 27 glyph symbols; the two-glyph variant applies a second table to the same letter with the intermediate glyph as state. Spaces after the glyphs y, n, r, l, m at a minimum token length of three and a maximum of nine. Independent sentences per line.

Naibbe cipher. Tables from the file naibbe_tables.csv of the author's repository (138 unigram strings, 138 prefix strings, 138 suffix strings for 23 letters and six tables). Preprocessing: lower case, $w \rightarrow uu$, $j \rightarrow i$, $k \rightarrow c$, letters only. Tokenization of the plaintext into one- or two-letter tokens with unigram probability 17/36 (the last letter of a line is always a unigram). Deck: 20 alpha, 8 beta1, 8 beta2, 8 beta3, 4 gamma1, 4 gamma2, shuffled and dealt without replacement, reshuffled when exhausted; one card per plaintext letter (variant: one card per token). Two-letter tokens are prefix(first letter) + suffix(second letter). Post-processing: each ciphertext space is removed with probability 0.03 (variant: none). Plaintext: Latin (Caesar) or Italian (Dante); one independent

sentence per line (variant: running text). State-dependent variant: with probability w the table of the next token is drawn from a fixed random distribution over the six tables conditioned on the last glyph of the previous token (one Dirichlet(c) draw per glyph), otherwise from the deck.

S4. Note on uncertainty estimation

A page-level bootstrap with replacement was tried first and rejected: duplicated pages create identical lines and token pairs, which inflates the plug-in mutual information (token MI interval 0.49–0.71 around a point estimate of 0.08) and depresses the hapax share. The leave-one-quire-out jackknife (Table 10) has no duplication and is the uncertainty measure used throughout. The jackknife standard errors of the excess-MI statistics include the variation of the estimator's residual bias with sample size.